

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I **SUDHARSAN D/192524063** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SUDHARSAN D

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer

encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File Size = **150 MB**
- Bandwidth = **50 Mbps**
- Segment Size = **1500 bytes**
- Header + Trailer = **40 bytes/frame**

Calculation:

- File size = **150,000,000 bytes**
- Number of Segments = $150,000,000 \div 1500 = \mathbf{100,000 \text{ segments}}$
- Number of Frames = **100,000**
- Total Overhead = $100,000 \times 40 = \mathbf{4,000,000 \text{ bytes (4 MB)}}$
- Total Data = **154 MB**
- Transmission Time = $(154 \times 8) \div 50 = \mathbf{24.64 \text{ seconds}}$

Answer:

- **Segments = 100,000**
- **Frames = 100,000**
- **Overhead = 4 MB**
- **Transmission Time = 24.64 s**

Explanation:

The **Transport Layer** divides the file into segments and provides sequencing,

acknowledgements, flow control, and error recovery. The **Data Link Layer** encapsulates each segment into frames, adds headers/trailers, performs error detection using CRC, and ensures reliable node-to-node communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Window Size = **6**
- Total Frames = **48**
- Frame Size = **1024 bytes**

Calculation:

- Number of Windows = $48 \div 6 = \mathbf{8 \text{ windows}}$
- One frame lost in each window
- Retransmissions = **8 frames**

Answer:

- **Transmission Windows = 8**
- **Retransmissions = 8**

Explanation:

Flow control prevents the sender from overwhelming the receiver. The sliding window allows multiple frames to be transmitted before waiting for acknowledgements, improving throughput while ensuring reliable communication.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- Packet Size = **12,000 bytes**
- MTU = **1500 bytes**
- IP Header = **20 bytes**

Calculation:

Maximum Data per Fragment

$$= 1500 - 20$$

$$= \mathbf{1480 \text{ bytes}}$$

Fragments

$$= \text{Ceiling } (12000 \div 1480)$$

$$= \mathbf{9 \text{ fragments}}$$

Header Overhead

$$= 9 \times 20$$

$$= \mathbf{180 \text{ bytes}}$$

Answer:

- **Fragments = 9**
- **Header Overhead = 180 bytes**

Explanation:

The **Network Layer** fragments packets whenever they exceed the MTU. Each fragment receives its own IP header containing identification, offset, and flags. The destination host reassembles all fragments correctly before delivering the original packet.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Calculation:

Window Size = **6**

Frames = **48**

Windows = $48 \div 6 = 8$

Lost Frames = **8**

Answer:

- **Windows = 8**
- **Retransmissions = 8**

Explanation:

Flow control synchronizes the sender and receiver speeds, preventing congestion and improving bandwidth utilization while ensuring reliable transmission through retransmissions.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- Video File = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Calculation:

Checkpoints before failure

$$390 \div 60$$

= **6 checkpoints**

Last checkpoint

= **360 MB**

Data to retransmit

390 – 360

= **30 MB**

Answer:

- **Checkpoints = 6**
- **Retransmitted Data = 30 MB**

Explanation:

The **Session Layer** establishes, manages, and terminates communication sessions. Synchronization checkpoints enable transmission to resume from the last saved point after a failure, reducing retransmission time and improving communication efficiency.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original File = **900 MB**
- Compression = **40%**
- Encryption Overhead = **5%**

Calculation:

Compressed File Size

$$= 900 \times (100 - 40)\%$$

$$= 540 \text{ MB}$$

Encrypted File Size

$$= 540 \times 1.05$$

$$= 567 \text{ MB}$$

Total Reduction

$$= ((900 - 567) \div 900) \times 100$$

$$= 37\%$$

Final Answer

- **Compressed File Size = 540 MB**
- **Encrypted File Size = 567 MB**
- **Overall Reduction = 37%**

Explanation

The **Presentation Layer** performs data compression and encryption. Compression reduces the amount of data transmitted, improving bandwidth utilization and reducing transmission time. Encryption protects data confidentiality and prevents unauthorized access during communication.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Data = **3 GB**
- Throughput = **120 Mbps**
- Request Size = **3 MB**

Calculation:

Convert Data

$$3 \text{ GB} = \mathbf{3000 \text{ MB}}$$

Number of Requests

$$3000 \div 3$$

$$= \mathbf{1000 \text{ requests}}$$

Convert Data into Megabits

$$3000 \times 8 = \mathbf{24,000 \text{ Mb}}$$

Transmission Time

$$24,000 \div 120$$

$$= \mathbf{200 \text{ seconds}}$$

Final Answer

- **Requests Generated = 1000**
- **Transmission Time = 200 seconds**

Explanation

The **Application Layer** provides services such as FTP for reliable file transfer. It

manages user authentication, file access, communication protocols, and error reporting, ensuring secure and efficient data exchange

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Application = **4 ms**
- Transport = **3 ms**
- Network = **5 ms**
- Data Link = **2 ms**
- Physical = **1 ms**
- Propagation = **18 ms**
- Transmission = **12 ms**

Calculation:

Processing Delay

$$= 4 + 3 + 5 + 2 + 1$$

$$= \mathbf{15\ ms}$$

Total Delay

$$= 15 + 18 + 12$$

= 45 ms

Final Answer

- **Processing Delay = 15 ms**
- **End-to-End Delay = 45 ms**

Explanation

Each OSI layer performs a specific task. The Application Layer handles user services, the Transport Layer provides reliable delivery, the Network Layer performs routing, the Data Link Layer handles framing and error detection, and the Physical Layer transmits bits over the medium. Together they ensure successful communication.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance. \

Given:

- Useful Data = **80 MB**
- Frame Size = **1600 bytes**
- Overhead = **80 bytes/frame**

Calculation:

Useful Data

80 MB = **80,000,000 bytes**

Frames Required

$$80,000,000 \div 1600$$

$$= \mathbf{50,000 \text{ frames}}$$

Total Overhead

$$50,000 \times 80$$

$$= \mathbf{4,000,000 \text{ bytes (4 MB)}}$$

Transmission Efficiency

Useful Data

$$= 80 \div (80 + 4)$$

$$\times 100$$

$$= \mathbf{95.24\%}$$

Final Answer

- **Frames = 50,000**
- **Overhead = 4 MB**
- **Efficiency = 95.24%**

Explanation

Protocol overhead consists of headers and trailers added by different OSI layers. Although overhead consumes bandwidth, it is essential for addressing, routing, synchronization, flow control, and error detection, ensuring reliable communication.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte**

segments, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original Record = **240 MB**
- Compression = **30%**
- Segment Size = **1400 bytes**
- Frame Overhead = **60 bytes**

Calculation:

Compressed File

$$240 \times 70\%$$

$$= \mathbf{168 \text{ MB}}$$

Compressed Data

$$168 \text{ MB}$$

$$= \mathbf{168,000,000 \text{ bytes}}$$

Segments

$$168,000,000 \div 1400$$

$$= \mathbf{120,000 \text{ segments}}$$

Protocol Overhead

$$120,000 \times 60$$

$$= 7,200,000 \text{ bytes}$$

$$= 7.2 \text{ MB}$$

Final Data Transmitted

$$168 + 7.2$$

$$= 175.2 \text{ MB}$$

Final Answer

- **Compressed File Size = 168 MB**
- **Segments = 120,000**
- **Protocol Overhead = 7.2 MB**
- **Final Data Transmitted = 175.2 MB**

Explanation

Multiple OSI layers cooperate to ensure secure and reliable communication. The **Presentation Layer** compresses data to reduce transmission size, the **Transport Layer** segments the data and ensures reliable delivery, the **Data Link Layer** frames the data and performs error detection, and the **Physical Layer** transmits the bits through the communication medium. Together, these layers provide efficient, accurate, and secure transmission of sensitive medical records.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

and
coherence

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand